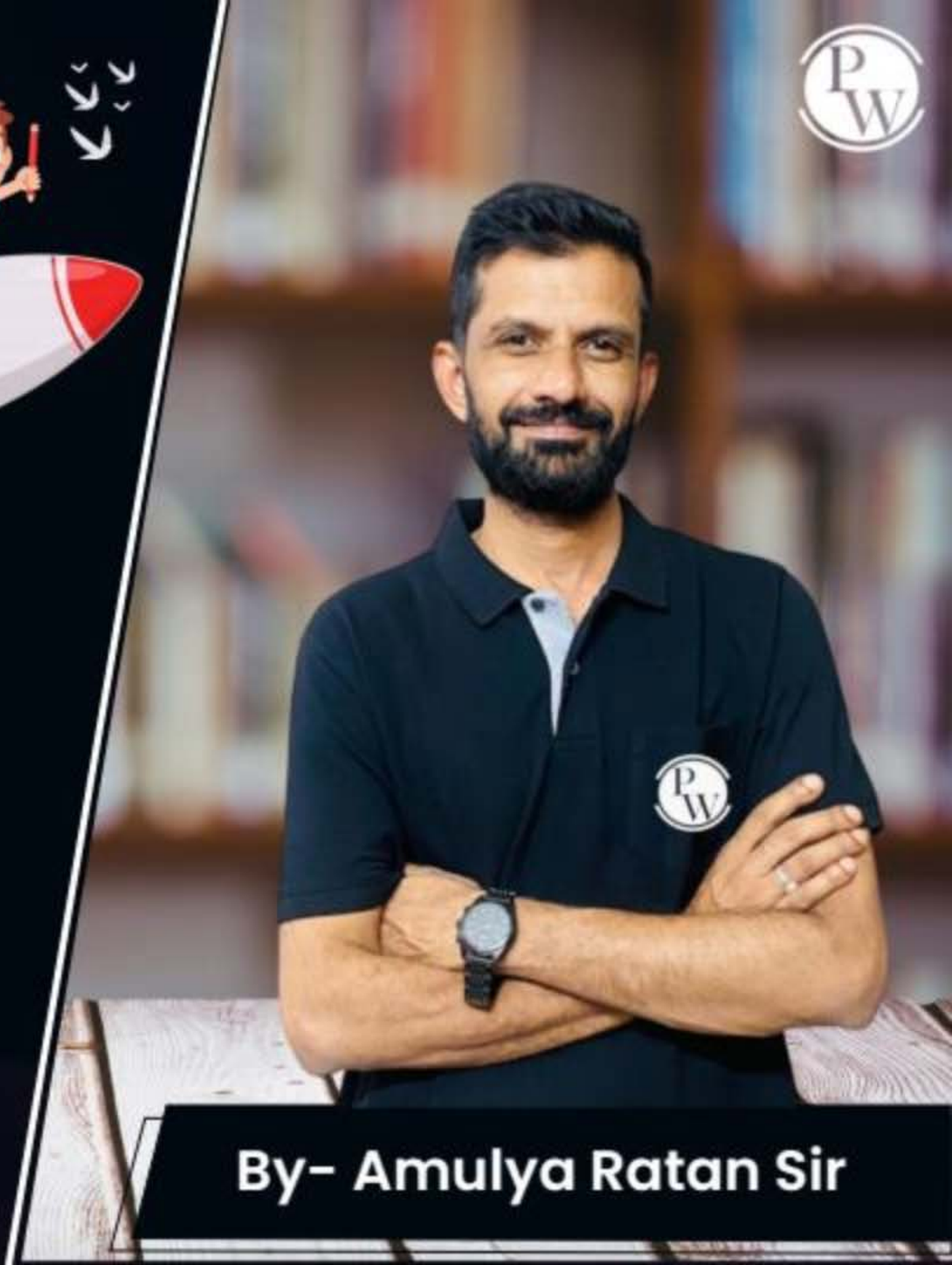
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 06

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Recap of Previous Lecture

Topic Mixtures Alligations

Topics to be Covered

Topic-1 Number System

[MCQ]

C.P of each kg $= \frac{300}{5} = 60$

#Q. 5kg of ghee was bought by Vinod for ₹300. One kg from spoilt. He sells the remaining in such a way that on the whole he incurs a loss of 10%. At what price per kg does he sell the ghee?

Assignment

$\frac{x}{60} \times \frac{4}{5} = \boxed{0.9}$

$\Rightarrow 4x = 0.9 \times 300$

$x = \frac{270}{4} = 67.5$

A ₹ 46.25

B ₹ 45.70

C ₹ 67.50

D ₹ 46.60

[MCQ]

#Q. How many litres of water must be added to 16 litres milk containing 10% water to dilute it with 20% water?

Mix : water
= 80 : 10
= (8 : 1) ×2
= 16 : 2

Assignment

Mix Water

10% 100%

20%

80 10

(A) 3 litres

(B) 2 litres

(C) 1 litre

(D) 4 litres

[MCQ]

#Q. A mixture contains milk and water in the ratio 5:1. On adding 5 liters of water, the ratio of milk to water becomes 5:2. the quantity of milk in the mixture earlier was:

A. 16 litres

B. 25 litres

C. 32.5 litres

D. 22.75 litres

30 lit

5 lit

Mix

Water

$\frac{7}{42} = \frac{1}{6}$

$1 = \frac{42}{42}$

$\frac{2}{7} = \frac{12}{42}$

$\frac{30}{42}$

$\frac{5}{42}$

Mix : water

$= 30 : 5$

$\frac{5}{6} \times 30 = 25$ lit

Assignment

[MCQ]

#Q. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

A. 3:8

B. 5:3

C. 7:3

D. 4:5

A B

15 20

16.5

3.5 1.5

A : B

= 3.5 : 1.5

7 : 3

Assignment

[MCQ]

A : B = 2 : 10

1 : 5

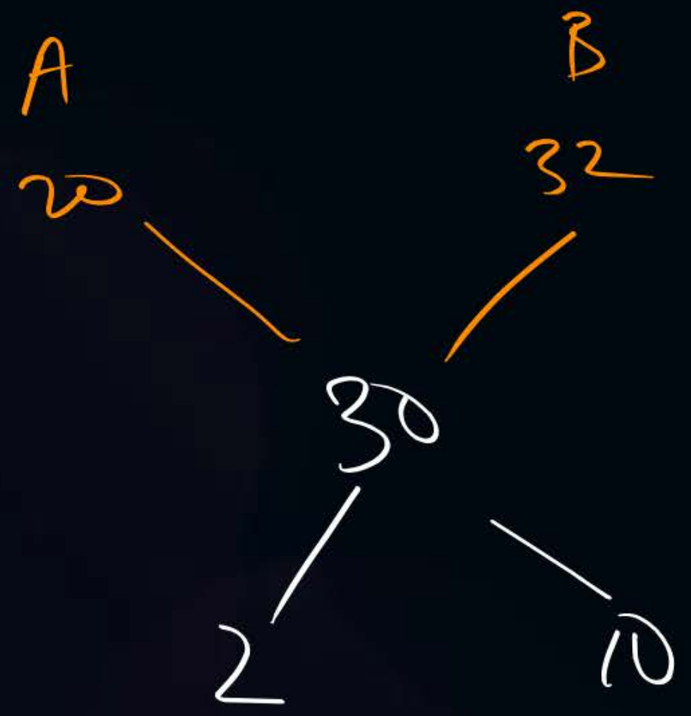
#Q. In what ratio must Ankit mix two varieties of sugar worth ₹20 per kg and ₹32 per kg so that by selling the mixture at ₹36 per kg he may gain 20%?

A. 1:5

B. 3:5

C. 2:3

D. 3:2



Assignment

S.P = 36

P% = 20%

$\frac{36}{C.P} = 1.2$

$\frac{36}{1.2} = C.P$

Basics of Number System

Infinity

Natural

7) 0 (0
0

'∞'

0) 9 (

3) 15 (5
15

2) 16 (8
16

$\frac{15}{3} = 5$

$\frac{16}{2} = 8$

$\frac{0}{7} = 0$

$\frac{9}{0} = \infty$

Undefined

Integers

Whole Numbers

$0, 1, 2, 3, 4 \ldots\ldots \infty$

$\ldots\ldots \infty$

$-4, -3, -2, -1, 0, 1, 2, 3, 4 \ldots\ldots$

Even number

∞

Odd Number

$2 \overline{)3}\,(1$

$\underline{2}$

1

ODD Number

Rational

$5 = \frac{5}{1}$ ✓

$-8 = \frac{-8}{1}$ ✓

$3.8 = \frac{38}{10}$ ✓

$0 = \frac{0}{1}$ ✓

$42.\overline{4}$

$= \frac{382}{9}$ ✓

$$10x = 424.444\ldots$$

$$-\quad x = 42.444\ldots$$

$$9x = 382$$

$$x = \boxed{\frac{382}{9}}$$

Irrational

$\sqrt{2} = 1.4142135623 9734018 \ldots$

$\sqrt{3} =$

Irrational

$\pi = \frac{22}{7}$ OR 3.14 → Rational

Decimals

- Terminating
- Non-Terminating
  - Recurring
  - Non-Recurring

[MCQ]

#Q. Find the answer of given expression: 1 - 2 + 3 - 4 + 5.....2024.

-1 -1 -1

-1012

[MCQ]

#Q. What would the unit digit in the answer of given expression:

$1! + 2! + 3! + \ldots\ldots\ldots\ldots 2024!$

$1 + 2 - 6 + 24 - 120 + 720 - \ldots$

-3 120 $+21$ $= 1$

$1 - 3 + 6 - 24 + 120 - 720$

-2 $+4$ 0

$\boxed{13}$

$1! = 1$

$2! = 2$

$3! = 6$

$4! = 24$

$5! = 120$

$120 \times 6 = 720$

$720 \times 7 = 5040$

[MCQ]

2003 → 2012

#Q. What would the remainder when the answer of given expression is divided by 3?

$1! + 2! + 3! + \ldots\ldots\ldots\ldots\ldots 2024!$ = Ans

$1! \Rightarrow 1$

$2! \Rightarrow 2$

$3! \Rightarrow 6$

$4! \Rightarrow 24$

3) Ans (

R = 0 ?

(A) 1

(B) 0

(C) 2

(D) Can't be determined

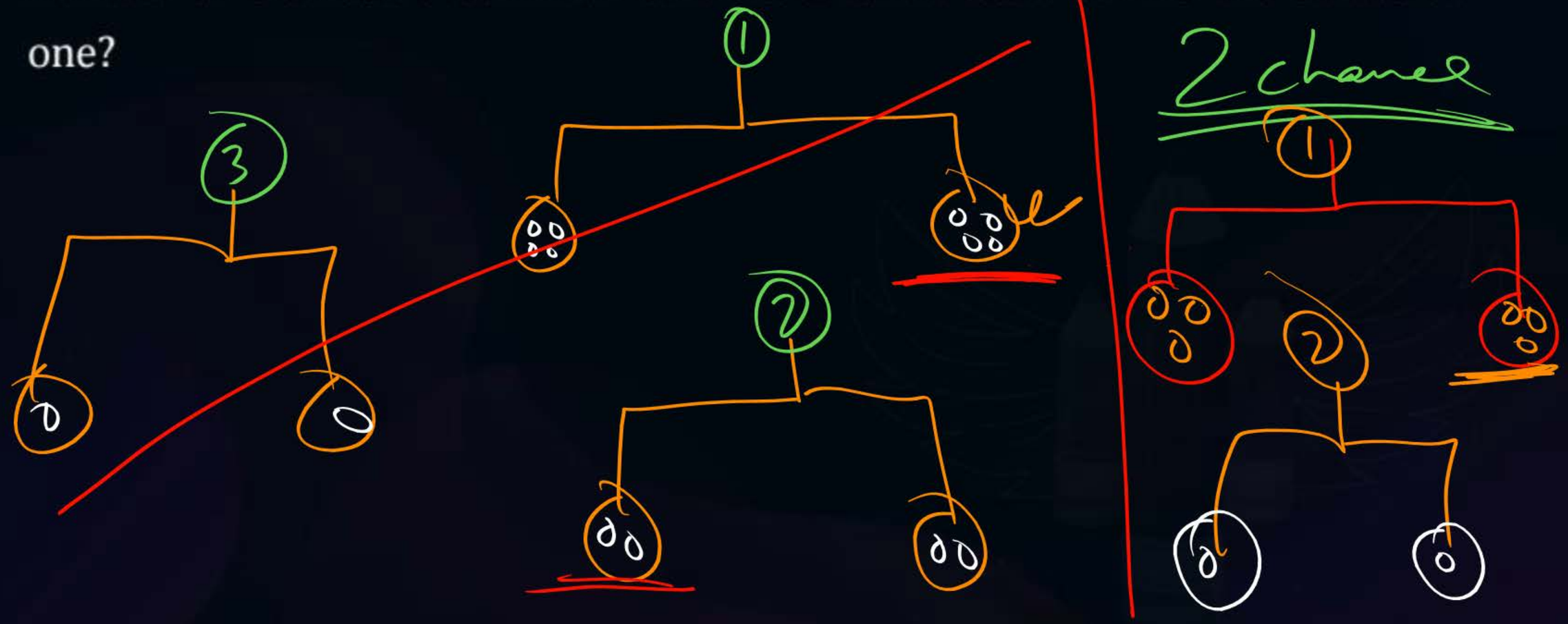
[MCQ]

~~7 balls~~ 6 balls

Case 1 =

Case 2 ≠

#Q. There are 8 balls of which one is defective. Given that the defective ball is of less weight and remaining are of equal weights. What are the minimum number of chances a common balance is to be used to find the defective one?



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Q. Due to reduction of 10% in the rate of sugar, a housewife can buy 5 kg more sugar for ₹ 600. Find the rate of sugar per kg after reduction.

$$0.9 \times \frac{600}{x} = \boxed{\frac{600}{x+5}}$$

[MCQ]

#Q. A bag consist of 48 red colour balls, 16 green colour balls, 12 yellow colour balls, 11 black colour balls and 11 white colour balls. How many minimum number of balls are to be taken out from the bag randomly so that we get at least two balls of same colour?

blind folded

6

[MCQ]

#Q. A shelf consist of 40 socks. 40% of these are black and remaining are white. How many minimum number of socks are to taken out from the shelf randomly (blindly), so that we get atleast two white socks?

40

Black 16

White 24

18

25

[MCQ]

#Q. A box consist of 40 pairs of shoes of equal size. 40% of these are black and remaining are white. How many minimum number of shoes are to taken out from the box randomly, so that we get at least a pair of black shoes?

48
+ 16
+ 1
65 ie

80

Black 32

White 48

2 black

50

[MCQ]

#Q. How many zeroes would be at the end in the answer of 50! ?

Assignment

A 12

B 15

C 23

D 24

2 mins Summary

Topic Number System

Basics

Logically

THANK - YOU